DevOps with AI

CSE636 · 1.5 Units · Fall 2025

California Science and Technology University

Lecturer: Qingsong Zhang · qingsong.zhang@cstu.edu

Revised for the era of Agentic AI

From AI-*assisted* to AI-*agentic* DevOps

> Students move beyond using AI as a passive assistant to designing **agentic systems** that perceive, plan, use tools, and act across the software delivery lifecycle — with guardrails and human oversight.

DevOps → DevSecOps → AIOps → **Agentic DevOps**

Course Description

This course explores how AI, ML, and **autonomous AI agents** are integrated into DevOps to automate, optimize, and enhance software development and operations.

Hands-on experience with:

- AI-assisted & **agentic CI/CD** pipelines
- AI **coding agents** and the **Model Context Protocol (MCP)**
- Predictive analytics & forecasting
- Intelligent monitoring & observability
- Autonomous incident response (**agentic SRE**)
- AI **security and governance**

Learning Objectives

By the end of the course, students will be able to:

1. Explain how AI agents & ML augment DevOps across **levels of autonomy** (assistant → human-in-loop → human-on-loop → autonomous).
2. Design & integrate **agentic automation** into CI/CD pipelines, incl. MCP-connected tools.
3. Use **predictive analytics & forecasting** for reliability, performance, and cost (FinOps).
4. Build AI-/agent-driven **monitoring, RCA, and autonomous incident response**.
5. **Evaluate** AI agents, frameworks & protocols at scale — with security & governance guardrails.

Course & Faculty Information

Lecturer	Qingsong Zhang
E-mail	qingsong.zhang@cstu.edu
Time	Wed 7:30–9:00pm (Online) · Sat 9:00–10:30am (Onsite)
Contact / Credit	23 Hours · 1.5 Units
Office hours	By Appointment
Prerequisite	Python; basic computing; Git & Docker recommended. No prior ML/LLM experience required.

Student Learning Outcomes

Assessed and reinforced in this course (not limited to):

- **Communication**
- **Critical Thinking**
- **Information Literacy**
- **ML, AI, and AI-agent / agentic-systems fundamentals**

> **Responsible AI-tool use:** students are expected to *use* AI agents in labs — and to **disclose and critically evaluate** their use.

Textbook & References

Supplementary text: *Mastering ChatGPT and Prompt Engineering*

Current references reflecting agentic DevOps:

- Anthropic — *Building Effective Agents* · Claude Code docs
- **Model Context Protocol (MCP)** spec & SDKs
- OpenAI Platform docs · Google Agent Development Kit (ADK)
- **OpenTelemetry GenAI** semantic conventions
- DORA metrics · Platform Engineering (Backstage) · Open Policy Agent

Evaluation & Grading

Component	Weight
Case Discussions	30%
Group Project	20%
Final Exam	20%
Mid-term Exam	15%
Class Participation	15%
Total	**100%**

A+ 98–100% · A 93–97% · A- 90–92% · B+ 88–89% · B 83–87% · B- 80–82% · C+ 78–79% · C 73–77% · C- 70–72% · D+ 68–69% · D 63–67% · D- 60–62% · F < 60%

The 7-Week Journey

Foundations → Tooling → Pipelines → Prediction
→ Observability → Incident Response → Platform & Governance

13 sessions · weekly labs · weekly assignments · capstone project

Week 1 — Foundations of AI-Assisted & Agentic DevOps

S1 · From DevOps to Agentic DevOps

- Evolution & why agents matter now; assistants vs. agents vs. autonomous
- Anatomy of an agent: perception, planning, tools, memory, act–observe loop
- Case studies: Claude Code, Copilot agent mode, Cursor, Devin

S2 · AI/ML & LLM Foundations

- LLM capabilities: reasoning, tool/function calling, structured output
- DevOps data as agent context; **RAG**; context engineering; connecting via **MCP**

Lab: cloud DevOps lab + run a coding agent on a sample repo · **Assignment:** report on 3 agentic-DevOps deployments

Week 2 — AI Agent Tooling, Protocols & Platforms

S3 · AI Coding Agents & Agentic Tooling

- Claude Code, Cursor, Copilot, Devin, Codex; AIOps platforms (Datadog, New Relic, Dynatrace, PagerDuty)
- Capabilities, weaknesses, and **when *not* to use an agent**

S4 · Agent Protocols, Frameworks & Environments

- **MCP** servers/tools/resources; LangGraph, CrewAI, AutoGen, ADK
- Cloud agent services; **agent permissions** (least privilege, sandboxing)

Lab: pipeline that invokes an agent + connect an MCP server · **Assignment:** compare frameworks + governance plan

Week 3 — Agentic CI/CD Pipelines

S5 · AI & Agents in Code Quality and Testing

- Agentic code review & static analysis; AI test generation, self-healing tests
- **Eval harnesses for AI-generated code**

S6 · Self-Optimizing & Self-Healing Pipelines

- Agents that fix failing builds & open PRs; predictive build-failure detection
- **Guardrails on autonomous merges**: approval gates, blast-radius limits

Lab: agent detects a failing build → proposes fix → opens PR (gated) · **Assignment:** CI pipeline that auto-remediates a failure class

Week 4 — Predictive Analytics & Capacity Intelligence

S7 · Risk Prediction & Deployment Intelligence

- Predicting failed deployments; canary/blue-green with **AI gating**
- Agentic rollback decisions; deployment success scoring

S8 · Capacity & Performance Forecasting

- Forecasting infra usage; **FinOps & cost optimization**
- ML-driven autoscaling (predictive HPA, KEDA); time-series forecasting

Lab: train a time-series model for CPU/memory → autoscaling decision · **Assignment:** AI forecasting for K8s autoscaling + cost

Week 5 — Intelligent Monitoring, Observability & Agent Telemetry

S9 · AI-Driven Anomaly Detection

- ML on logs/metrics/traces; isolation forests + LLM-based detection
- **Observability *for* agents**: OpenTelemetry GenAI conventions — tokens, cost, latency

S10 · Root Cause Analysis with AI Agents

- Dependency mapping; log/trace correlation with LLMs; alert grouping
- **Agentic RCA**: investigate-and-report agents

Lab: anomaly detector + instrument an agent with OTel GenAI · **Assignment:** anomaly detection → AI root-cause summary

Week 6 — Autonomous Incident Response & Agentic SRE

S11 · Self-Healing Systems & Agentic SRE

- AI as a "virtual SRE"; autonomous rollback/failover with **guardrails**
- Agent reasoning loops (ReAct, reflection); **blast-radius control**

S12 · Alert Triage, Runbooks & ITSM Integration

- Intelligent suppression & correlation; **agentic runbooks**
- PagerDuty / ServiceNow / Opsgenie; AI-assisted postmortems

Lab: Python agent (MCP tools) triages a simulated incident → gated remediation · **Assignment:** self-healing microservice with approval step

Week 7 — Agentic IaC, Platform Engineering, Security & Governance

S13 · Agentic IaC, Platform Engineering & AI Governance

- Agents for Terraform/OpenTofu/CloudFormation; **Policy-as-Code (OPA)**
- Internal Developer Platforms (Backstage) & golden paths
- **Agent security**: prompt injection, permission scoping, sandboxing, secrets
- Supply-chain security (SLSA), audit trails, responsible-AI governance

Capstone: agent-augmented IaC + end-to-end agentic DevOps project (pipeline → deploy → observe → auto-remediate, with guardrails)

Academic Integrity & AI Tools

- All submitted work must be **original** or **properly cited**.
- AI coding agents/assistants are **permitted and expected** in designated labs — but must be **disclosed**.
- Students remain **fully responsible** for correctness, security, and originality of all submitted work, including AI-generated content.
- Using AI where prohibited (e.g., exams) is a policy violation.

Welcome to CSE636

Let's build the agentic future of DevOps

Questions? Office hours by appointment · qingsong.zhang@cstu.edu